INDEPENDENT EVALUATION

GSA CALC+ API Testing Record

GSA MAS ceiling rates skill validated across eight end-to-end runs on two Claude models; Rounds 2 and 3 patches shipped

Skill

gsa-calc-ceilingrates (post-Round 3 patches; 112/112 Wave 1 assertions passed)

github.com/1102tools/federal-contracting-skills

Date

April 2026

1102tools.com

Part 1: For Federal Acquisition Users

The bottom line

Independent testing in April 2026 (8 end-to-end runs, 112 binary assertions graded, 2 Claude models) shows the GSA CALC+ skill reliably pulls MAS ceiling rate distributions across four real-world federal acquisition workflows. Wave 1 testing surfaced a critical silent-wrong-answer bug in the skill's response schema documentation that was patched before publication. Post-patch validation on a second model confirmed the fix held. Round 3 patches shipped in April 2026 after findings surfaced during LH/T&M IGCE testing.

Wave 1 aggregate: 112/112 (100%). Round 3 patches shipped.

Scenarios tested and how reliably they work

Workflow	Models	Result
Baseline rate distribution (single LCAT keyword, full percentile pull)	Opus, Sonnet	Reliable
Keyword refinement via suggest-contains for noisy terms (Program Manager)	Opus, Sonnet	Reliable
SIN-filtered query with clearance layer (HACS, 54151HACS)	Opus, Sonnet	Reliable
Rate validation workflow ($185/hr Senior Software Developer, IGCE narrative)	Opus, Sonnet	Reliable

Manual-verification checklist

Scan every CALC+ output for these items before using in a contract file:

1. Confirm the aggregation path was used correctly. The skill's canonical statistics all live under `response["aggregations"]`, NOT at the top level. Pre-patch the skill documented them as

top-level which caused silent `None` returns. If an output contains `None` or zero values where you expect real statistics, the model likely read from the wrong path.

2. `keyword=` is for scoping only, not rate stats. The `keyword=` parameter searches three fields simultaneously (labor_category, vendor_name, idv_piid). For rate analysis, use `suggest-contains=labor_category:<term>` to find real LCAT bucket names, then `search=labor_category:<exact value>` for stats. Outputs citing keyword-query rates in a price reasonableness memo should be flagged.

3. `security_clearance:yes` is a floor, not a true count. The filter does fuzzy partial-matching on variant bucket keys (`true`, `Yes`, `SECRET or TS/SCI eligible`). Counts are floors. For cleared requirements, the all-clearance HACS or cyber pool is often the more defensible benchmark than the cleared-filtered subset.

4. Record counts above 10,000 require the aggregation count. `hits.total.value` caps at 10,000 with `relation: "gte"`. Use `response["aggregations"]["wage_stats"]["count"]` for the true population size. A memo citing "10,000 records" when the true population is 45,000 misrepresents sample size.

5. SIN filter is case-sensitive. `54151HACS` works; `54151hacs` returns zero. Check SIN casing in the output.

6. Apply `price_range:30,500` by default for professional services IGCE work. CALC+ contains sentinel values ($9,999 placeholders, clerical floor rates) from messy vendor PPT submissions. The default bound excludes these. Adjust for specialty work (SCIF `50,600`; administrative `15,200`). Document the bound used in the memo.

7. `median_price` vs `histogram_percentiles["50.0"]`. Both present a "median" but use different interpolation. `histogram_percentiles["50.0"]` is the canonical median (linear interpolation between ranked values). If an output cites $1-$2 different from what the web UI shows, this is likely the cause; cite the histogram_percentiles value.

8. Dual-pool analysis for senior LCATs. For a "Senior X" requirement, both `suggest-contains=labor_category:Senior X` (title-match) and `X + min_years_experience:8` (experience-match) are legitimate benchmark pools. They often differ by 10-15% at median. The CO decides which best matches the vendor's LCAT description. An output citing only one pool should be flagged.

Choosing between Opus 4.7 and Sonnet 4.6

Short answer: both work. Use either. Post-patch behavior is effectively identical.

Opus 4.7 tends toward comprehensive workflow execution. In testing, Opus provided 4-pool comparisons for rate validation, dual SIN cross-checks for HACS work, and more detailed narrative context. Opus's comparisons were slightly more thorough but no more defensibly accurate than Sonnet's.

Sonnet 4.6 is tighter in output and frequently uses exact-match searches over keyword queries. Sonnet S2 and S4 in testing produced visualization-rich output (ASCII percentile positioning, comparison tables) that reads well in a CO memo. Sonnet's exact-match discipline is methodologically slightly tighter than Opus's keyword-based approach.

Neither model hits tool-use limits on CALC+ queries. Both skills complete in a single response.

What the skill does not do

- **It does not reflect task-order-level prices paid.** CALC+ contains awarded NTE (Not-To-Exceed) ceiling rates from master MAS contracts. Order-level actuals are typically 10-20% below ceiling per FAR 8.405-2(d).
- **It does not apply geographic cost adjustment.** Rates are worldwide. A cleared DC-area vendor and an uncleared Midwest vendor may appear in the same bucket.
- **It does not substitute for a price reasonableness determination.** The CO still makes the determination per FAR 15.4. CALC+ data informs it; it does not replace it.
- **It has not been tested on** multi-vendor competitive range analysis, vendor-specific rate card queries for IGCE comparison, historical rate trending across contract years, or LCAT-to-LCAT comparison across multiple SINs.

Environmental gotchas on claude.ai web chat

Gotcha	What happens	Workaround
`aggregations` silently returns None if read from wrong path	Pre-patch skill documented paths as top-level	Fixed in Round 2 patch; current skill documents nested path explicitly
`suggest-contains` returns exactly 100 buckets	Default cap; silent truncation on noisy terms	Narrow the discovery term if you hit exactly 100
Keyword contamination understatement	Skill's anti-pattern language may overstate cross-field contamination on specific terms	For terms like "Program Manager" the 100+ long tail of LCAT variants dominates; always verify with `sum_other_doc_count` check

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Part 2: For Developers and Technical Reviewers

Testing methodology

Scenarios

Four scenarios selected before any testing began, chosen to exercise distinct CALC+ workflows:

- **S1 — Baseline reachability + aggregation path:** Full rate distribution for Information Security Analyst. Tests endpoint, parameter name (`keyword` vs `q`), aggregation path (`histogram_percentiles.values["50.0"]` vs top-level), distribution presentation.
- **S2 — High-ambiguity keyword refinement:** Program Manager via `suggest-contains` then `search=labor_category:<exact>`. Tests search refinement, large-result-set handling, discovery discipline.

- **S3 — SIN-filtered cleared cyber:** Rates under SIN 54151HACS with clearance layer. Tests SIN filter syntax, cleared LCAT data availability, filter+keyword combo, `security_clearance` filter behavior.
- **S4 — Rate validation workflow:** $185/hr Senior Software Developer reasonableness check. Tests real IGCE decision-support, percentile positioning, divergence analysis, narrative production.

Each scenario had a 14-point binary assertion matrix: 8 general (G1-G8, same across all scenarios) and 6 scenario-specific. Assertions written before any worker output was seen.

Environment

- claude.ai web chat, fresh conversation per run
- Skills installed: `gsa-calc-ceilingrates` with reference skill merged into main SKILL.md (Round 2)
- Models: one run on Opus 4.7, one run on Sonnet 4.6, per scenario
- Total: 4 scenarios × 2 models = 8 runs

Grading

Grader (Claude Code session separate from any worker run) read only the worker's final response text. Workers not coached during runs. Each assertion graded binary pass/fail. Suspicious details noted even when the assertion passed.

Wave 1 results

Scenario	Opus 4.7	Sonnet 4.6
S1 Baseline distribution	14/14	14/14
S2 Keyword refinement	14/14	14/14
S3 SIN filter + cleared	14/14	14/14
S4 Rate validation	14/14	14/14
Total	**56/56 (100%)**	**56/56 (100%)**

Aggregate: 112/112 (100%).

Cross-model data cross-validation

Both models pulled identical raw data from the API, confirming the skill's endpoint and parameters work consistently:

- **S1 Information Security Analyst (unfiltered):** Both models N=402, median $128.01-$128.02 (rounding), mean $133.21.
- **S2 Program Manager (broad keyword):** Both models N=7,763, P50 ≈ $180.70.
- **S3 HACS SIN cleared cyber:** Both models N=1,148, median $150.07.
- **S4 Senior Software Developer:** Opus (keyword-based) N=145 median $151.66; Sonnet (exact-match) N=54 median $157.44. Different sample sizes reflect different methodological approaches (keyword vs exact match), both defensible. Documented as a "dual-pool" patch candidate.

Round 1 findings: 9 skill bugs surfaced

From Wave 1 Opus worker self-assessments (ordered by severity):

1. **Aggregations schema documentation wrong (CRITICAL).** Skill documented `wage_stats`, `histogram_percentiles`, etc. as top-level response fields. Actual API nests all aggregations under `response.aggregations.*`. Caused silent `None` returns. 3 of 4 Opus workers hit this and corrected in-flight.
2. **`keyword=` contamination across 3 fields.** The parameter searches `labor_category`, `vendor_name`, and `idv_piid` simultaneously. Skill did not flag this as an anti-pattern for rate statistics.
3. **"Discover first" was scoped to vendor names in Rule #2.** Should be labor_category first (noisier and more relevant to IGCE use case).
4. **`security_clearance:yes` is fuzzy, not normalized.** Skill claimed "API normalizes"; actually returns multiple variant bucket keys (`true`, `Yes`, `SECRET or TS/SCI eligible`). Counts with this filter are floors.
5. **`hits.total` caps at 10,000 buried under Rate Limiting.** Should be prominent in "reading results" context. Sanity checks comparing `hits.total.value` vs `wage_stats.count` will false-alarm on every query over 10K.
6. **`suggest-contains` default 100-bucket cap undocumented.** Silent truncation on noisy terms.

7. **`page_size=1` aggregation-only idiom undocumented.** Efficient pattern for distribution pulls was not called out.
8. **SIN filter case sensitivity undocumented.** `54151HACS` works; `54151hacs` returns empty.
9. **Filter encoding guidance missing.** Pipe-delimited OR and comma-ranges work but when to URL-encode vs not was unclear.

Patches shipped before Sonnet wave

All patches applied to `~/.claude/skills/gsa-calc-ceilingrates/SKILL.md`. The `gsa-calc-ceilingrates-reference` split-skill architecture was merged into the main SKILL.md in the same pass (mirrors the BLS OEWS merge).

1. **New "Response Shape" section at top** with full envelope tree showing nested `aggregations.*` paths. Canonical JSON paths for the 7 IGCE-relevant statistics.
2. **New `distribution_snapshot()` recipe** as the Quick Start. Returns count/min/max/mean/P25/P50/P75 in one call with the correct paths baked in.
3. **Critical Rule #2 rewritten: "Never use `keyword=` for rate statistics."** Explicit anti-pattern with the Program Manager contamination example.
4. **Critical Rule #3 rewritten: "Discover Labor Categories First."** Reframed from vendor-name scope to LCAT scope (higher IGCE value).
5. **Critical Rule #4 added: "`security_clearance` filter is fuzzy, not normalized."** Documented variant bucket keys, counts as floors, recommendation to use all-clearance pool for IGCE.
6. **Critical Rule #5 added: "`hits.total` caps at 10,000."** Moved from Rate Limiting to prominent context; explicit guidance on `wage_stats.count` as source of truth.
7. **Critical Rule #7 added: "Use `page_size=1` for aggregation-only queries."**
8. **Critical Rule #6 added: "Messy Field Values"** including SIN case sensitivity, LCAT discovery, vendor name discovery.
9. **Filter encoding guidance added** (pipe-OR, comma-ranges, when to quote vs not).
10. **`price_range:30,500` outlier-strip convention** documented as IGCE default with adjustment ranges for specialty work.
11. **New `price_reasonableness()` dual-pool recipe** for senior LCATs (title-match vs experience-match pools).
12. **New `tiered_rate_card()` recipe** for seniority stratification.

13. **`median_price` vs `histogram_percentiles["50.0"]` interpolation difference documented.**
14. **`suggest-contains` 100-bucket cap documented** with truncation warning.
15. **Quick Reference workflow section at the end** (discover → stratify → position → narrate).
16. **Expanded Troubleshooting table** with all silent-wrong-answer bugs.

Wave 1 Sonnet validated the patches

Sonnet wave executed after Round 2 patches shipped. All 4 Sonnet runs (56/56) used the patched response paths correctly from the first attempt: - No NullType crashes (pre-patch, Opus hit the bug on 3 of 4 runs) - All Sonnet workers used `aggregations.wage_stats.*` from the start - All applied `price_range:30,500` convention - All followed `suggest-contains` → `search=labor_category:<exact>` discipline - Sonnet S3 empirically tested and confirmed the SIN case sensitivity patch

Truncation investigation

A targeted audit asked an Opus Wave 2 session to quote verbatim from three specific locations in the 719-line merged file: 1. Critical Rule #4 heading and first sentence (line 177) 2. `tiered_rate_card()` function signature (line 399) 3. Troubleshooting table entry for `wage_stats` returns None (line 678)

Worker returned all three verbatim with correct line numbers. **Truncation is not real web-chat behavior.** Workers who described mid-file sections as "truncated" were imprecise about "I didn't re-read the full file to check." The 719-line merged file loads fully. Same finding as BLS.

What was not tested

- Multi-vendor competitive range analysis
- Vendor-specific rate card queries (`vendor_rate_card()` recipe in skill)
- Historical rate trending across contract years
- LCAT-to-LCAT comparison across multiple SINs in one query
- The `exclude` parameter for outlier removal
- CSV export (`&export=y` parameter)

Round 3 patches shipped

Triggered by findings during LH/T&M IGCE Wave 1 testing. Three Opus workers building LH and T&M IGCEs against San Diego, Huntsville, and Dayton MSAs surfaced consistent usage patterns that the skill did not address cleanly. Patches shipped April 2026.

1. **`keyword=` vs `search=` rules clarified (Rule 2 rewritten).** Previous skill prohibited `keyword=` for rate statistics without distinguishing between formal rate analysis and directional sanity-layer validation in IGCE contexts. Rewritten to explicitly allow `keyword=` for: directional divergence band checks, corpus sizing, and discovery. `search=labor_category:<exact>` remains required for: specific vendor rate reasonableness (Workflow B), primary benchmarks in contract file output. Resolves contradiction between CALC+ skill and orchestration skills that use `keyword=` for sanity layer.

2. **Pre-flight endpoint check added before Quick Start.** New section at top of skill requires a single cheap query to verify HTTP 200 + `count` field presence before batching. Documents the canonical base URL (`https://api.gsa.gov/acquisition/calc/v3/api/ceilingrates/`) and lists historical URLs that no longer resolve (`calc.gsa.gov/api/v3/...` and `calc.gsa.gov/api/v1/rates/`). LH/T&M Scenario 2 and Scenario 3 workers hit 404s from stale URLs cited in an orchestration skill; the pre-flight pattern makes the failure mode unmissable.

3. **CALC+ corpus structural skew documented (Rule 2a added).** New rule tabulating keyword contamination direction and magnitude for common trap terms: `Program Manager` (+40% vs DoD/civilian PM wage, dominated by IT/cloud PMs at Tier 1 primes), `Analyst` (+20%, skews senior cyber/IT), `Engineer` (+15-30%, IT/cloud dominant), `Architect` (+30%), `Project Manager` (+15-25%). Lists underrepresented occupational families (logistics, DoD acquisition, physical engineering 17-2xxx, medical 29-xxxx, research lab 19-xxxx). Provides remediation protocol: narrow keyword → `suggest-contains` + `search=labor_category:<exact>` → thin-pool fallback language for N<30.

Round 4 patches queued

None block current ship state.

1. **Verify whether `suggest-contains` bucket cap is configurable via `size` parameter.** Currently unknown; skill notes the 100-bucket observation but has not tested expansion.
2. **Add `median_price` vs `histogram_percentiles["50.0"]` worked example.** Skill explains the interpolation difference in text; a numeric example on a known dataset would help

COs defend the chosen value.

3. **Add CO decision tree for pool selection.** When SOW language is title-weighted vs experience-weighted vs SIN-specific, which pool is the right anchor? Explicit flowchart would reduce methodology inconsistency across analysts.
4. **Document methodology choice between keyword and exact match for same LCAT.** Sonnet S4 exact-match got N=54; Opus S4 keyword search got N=145. Both defensible. Dual-pool workflow should explicitly address which to prefer in narrative when both are run.

Independent grading methodology

Wave 1 testing record produced under consistent methodology:

- Scenarios and assertion matrices committed in writing before any worker output was read
- Grader did not coach workers during runs
- Assertions graded strict on literal wording; ambiguous assertions noted and refined for the next wave (not retroactively reinterpreted)
- Methodology source: `calc-plus-wave1-runbook.md`
- All findings come from direct observation of worker output, not inference from memory of prior sessions

> *Testing record prepared April 2026 by James Jenrette / 1102tools. Independent grading methodology. MIT licensed. Source: github.com/1102tools/federal-contracting-skills.*